

3.2 PS1300

3.2.1 General information

The PS1300 system supply module uses 24 VDC industrial voltage to generate power for the electrically isolated X2X Link. This system supply module also provides 15 W of output power for additional stations on the bus.

- Electrical isolation of feed and X2X Link supply
- Protection with redundancy during parallel operation of multiple system supply modules
- Short circuit protection
- Overload protection

3.2.2 Order data

Model number	Short description	Figure
	System supply module	
X67PS1300	24 VDC X67 System supply, X2X Link supply 15 W, reverse polarity protection, short circuit protection, overload protection, parallel connection possible, redundancy operation possible, LED status indicators	
	Required accessories	
See 14 "Overview of pin connections" on page 647		

Table 110: PS1300 - Order data

3.2.3 Technical data

Product ID	PS1300
Short description	
System supply	Creates electrically isolated X2X Link supply
X2X Link supply input	
Rated voltage	24 VDC
Voltage range	18 - 30 VDC
Rated current	0.75 A
Fuse	Integrated
X2X Link supply output	
Rated output power	15.0 W
Overload behavior	Short circuit protection, overload protection
Parallel operation	Yes
Redundant operation	Yes, when input voltages are the same
General information	
Status indicators	Input voltage OK, output voltage OK
Electrical isolation Input - Output	Yes
Power consumption Internal	3.0 W
Connection type X2X Link supply input X2X Link supply output	M8 (4-pin) M12 (B-coded)
Certification Ex zone 2	CE, cRUus, GOST-R II 3G EEx nA II T5, IP67, Ta = 0 - 60°C
Operational conditions	
Operating temperature	0 to +60°C
Mounting orientation	Any
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of ambient temperature by 0.5°C per 100 m
Protection type	IP67
Storage and transport conditions	
Temperature	-25 to +85°C
Mechanical characteristics	
Dimensions (W x H x D)	53 x 85 x 42 mm
Weight	190 g
Torque for connections M8 M12	Max. 0.4 Nm Max. 0.6 Nm

Table 111: PS1300 - Technical data

3.2.4 Additional technical data

Product ID	PS1300
Output	
Rated voltage	20 VDC
Rated current	0.75 A
Supply failure bridging	>5 ms @ 24 VDC _{in} and I _{out} = 0.75 A
Efficiency	>80%
Isolation voltage between input and output	500 V _{eff}

Table 112: PS1300 - Additional technical data

3.2.5 Status LEDs

Figure	LED	Description
Status indicator 1: green  Status indicator 2: orange	Status indicator 1	Status indicator - X2X Link. The green LED is lit when the X2X Link supply is in the valid range.
	Status indicator 2	Status indicator for the module supply. The orange LED is lit when the module supply is in the valid range.

Table 113: PS1300 - Status LEDs

3.2.6 Connection elements

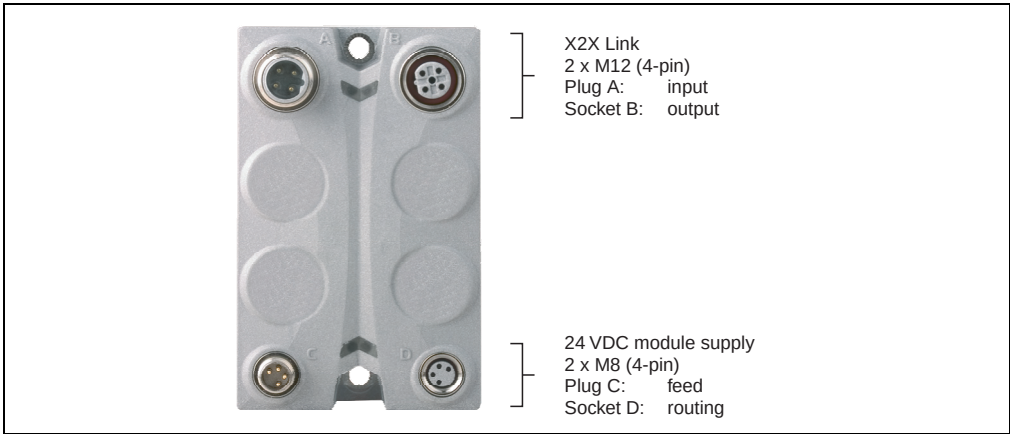


Figure 71: PS1300 - Connection elements

3.2.7 X2X Link

The PS1300 module is connected to the X2X Link with pre-assembled cables. The connection is made using a circular plug (2 x M12, 4-pin).

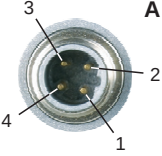
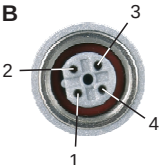
Connection	Pin assignments	
	Pin	Name
	1	X2X+
	2	X2X
	3	X2X _L
	4	X2X _I
	A ... B-coded plug in the module, input	
	B ... B-coded socket in the module, output	
	SHLD ... Shield connection made via threaded insert in the module	

Table 114: PS1300 - X2X Link

3.2.8 24 VDC module supply

The module supply connection is made using circular plugs (2 x M8, 4-pin). The supply feed is connected via plug C. Socket D is used for routing the supply to other modules.

The maximum permitted current for the circular plug is 8 A.

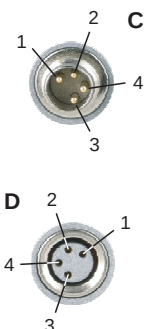
Connection	Pin assignments	
	Pin	Name
	1	24 VDC
	2	24 VDC
	3	GND
	4	GND
	C ... Plug on the module, supply feed D ... Socket on the module, supply routing	

Table 115: PS1300 - 24 VDC module supply